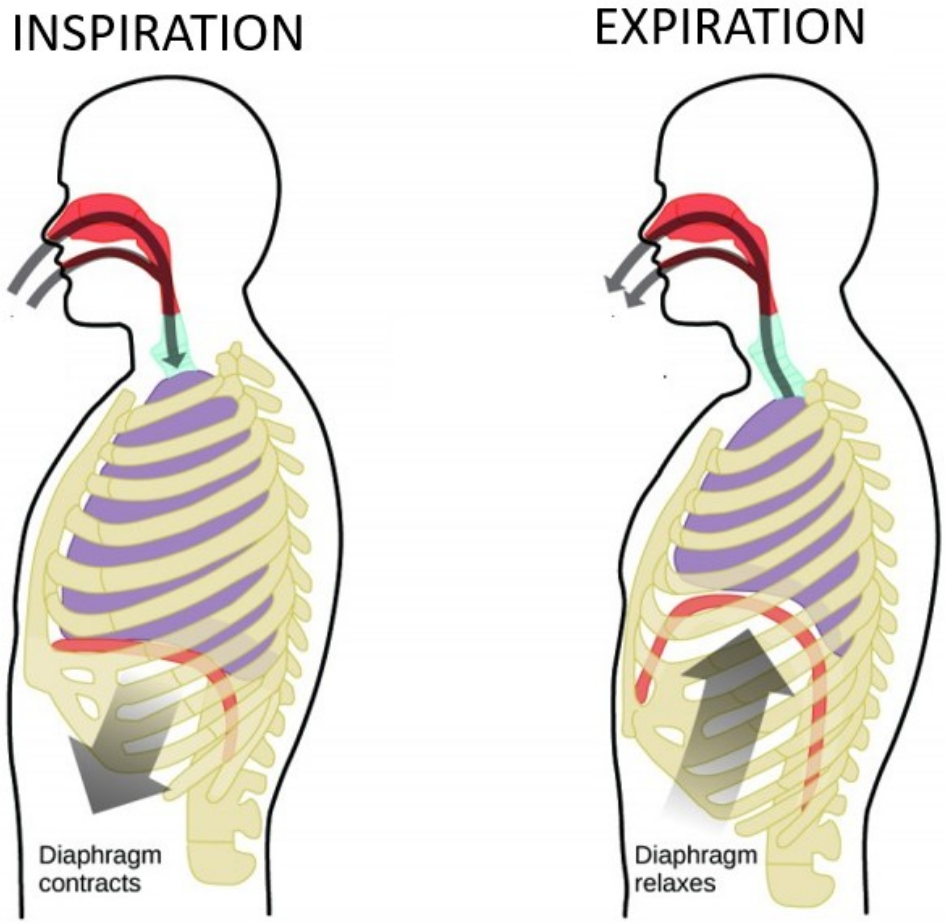
RESPIRATORY SYSTEM OSPE

DAY 3

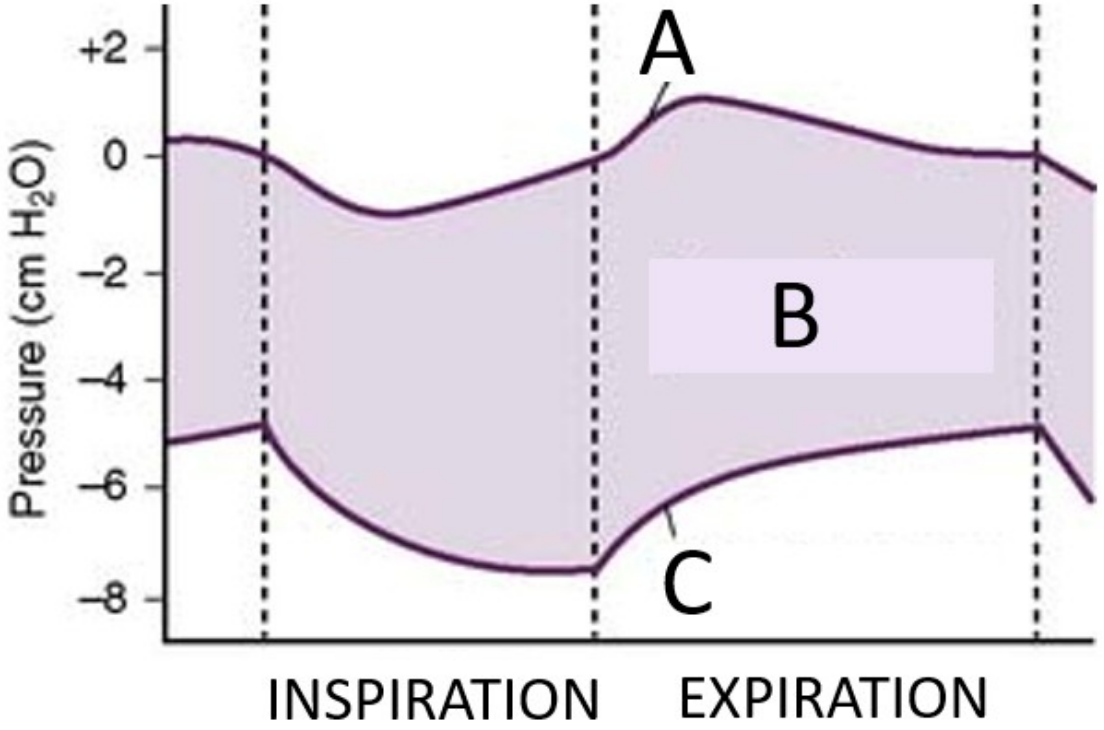
SPOT NO.1

1. Identify the physical law demonstrated by this diagram.
2. State the relationship between pressure and volume according to this law.
3. During inspiration, what happens to intrapulmonary pressure?
4. Name one muscle responsible for quiet inspiration.



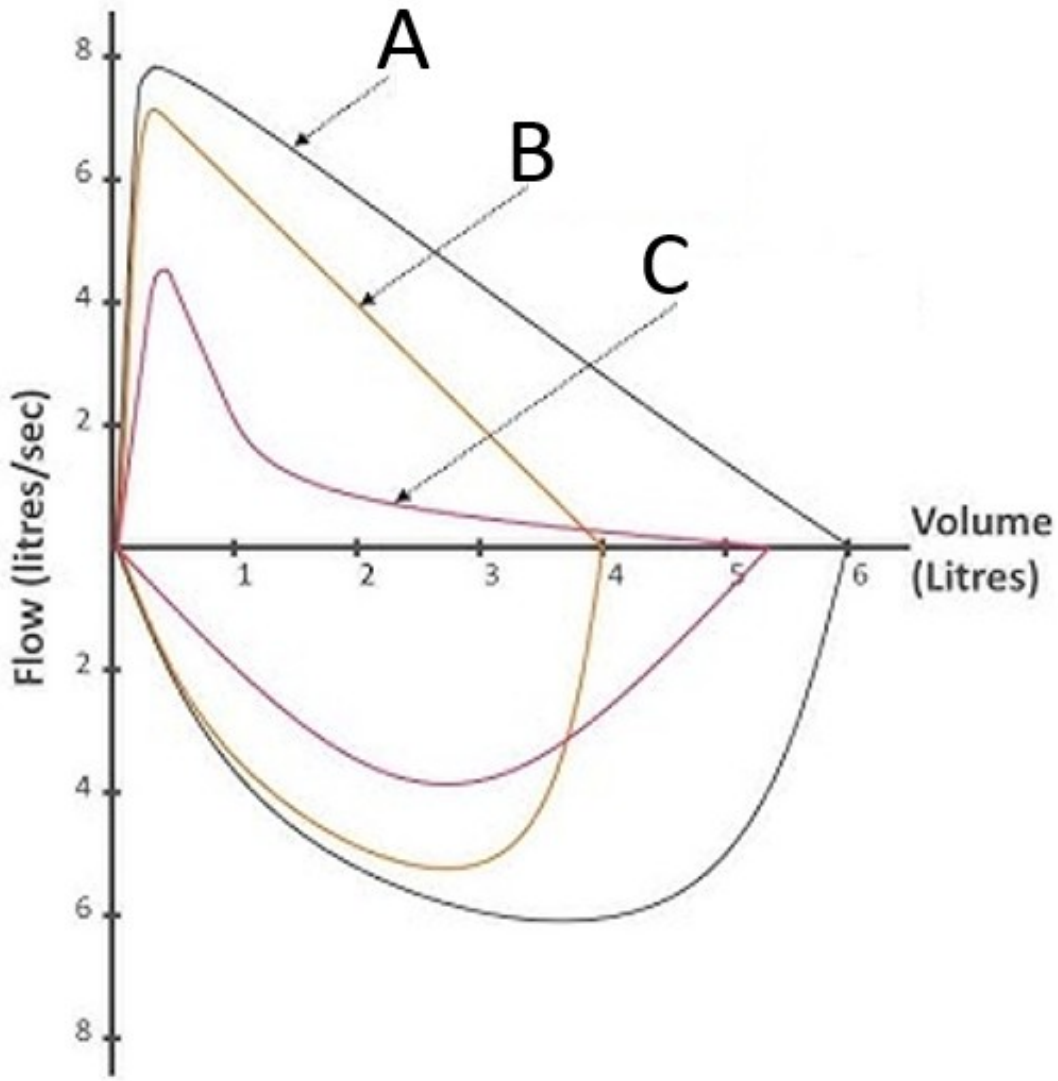
SPOT NO.2

1. Identify the pressures A, B, & C.
2. Define transpulmonary pressure.
3. What is its significance?
4. What will happen to transpulmonary pressure in pneumothorax?



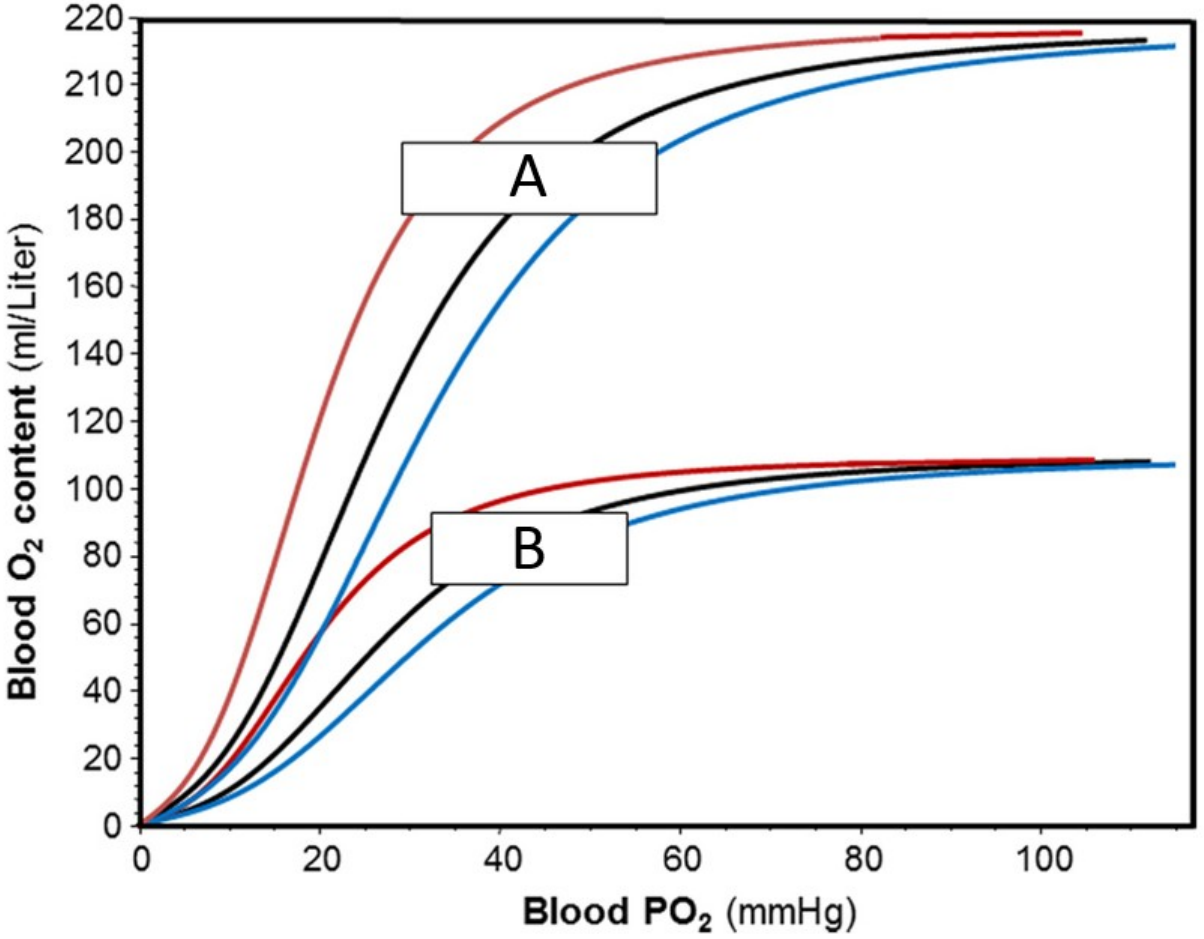
SPOT NO.3

1. Identify the type of test shown in the figure.
2. Which curve represents obstructive lung disease?
3. What happens to FEV_1/FVC ratio in obstructive disease?
4. Name one example of an obstructive lung disease.



SPOT NO.4

1. Identify the physiological disturbance shown in the diagram. (Curve B)
2. Name two common causes of hypoxia.
3. In which type of hypoxia is PaO_2 normal but O_2 content low?
4. Explain why oxygen therapy is ineffective in this type of hypoxia.



SPOT NO.5

Sea Level	760	100	40	60	98
10.000	523	60	31	29	87
18.000	380	38	26	12	72
22.000	321	30	22	8	60
25.000	282	7	4	3	9
35.000	179	0	0	0	0

1. Identify the physiological condition illustrated in the table.
2. What happens to alveolar PO_2 (PAO_2) with increasing altitude?
3. What is the main stimulus for increased ventilation at high altitude?
4. Name one compensatory change that occurs after acclimatization.